Tectonophysics, 33 (1976) 337—357

© Elsevier Scientific Publishing Company, Amsterdam — Printed in The Netherlands

PALEOMAGNETISM OF THE LOWER CAMBRIAN BRADORE SANDSTONES, AND THE ROTATION OF NEWFOUNDLAND

K.V. RAO and E.R. DEUTSCH

Geomagnetic Research Laboratory, Department of Physics, Memorial University of Newfoundland, St. John's, Newfoundland (Canada)

(Accepted for publication March 15, 1976)

ABSTRACT

Rao, K.V. and Deutsch, E.R., 1976. Paleomagnetism of the Lower Cambrian Bradore sandstones, and the rotation of Newfoundland. In: J.C. Briden (editor), Ancient Plate Margins. Tectonophysics, 33: 337—357.

Wegener proposed that Newfoundland has rotated 30 degrees anticlockwise relative to Labrador. We report here new paleomagnetic evidence bearing on this hypothesis, obtained from sandstones of the Lower Cambrian Bradore Formation on Canada Bay, Newfoundland. Isothermal remanence tests indicate that the entire natural remanence (NRM) of the samples resides in hematite. The mean NRM direction for 9 sites is $D = 153°$, $I = +52°$, $\alpha_{95} = 15°$, confirming R.F. Black's 1964 result from other Bradore exposures in northern Newfoundland, which he compared with rocks in New Brunswick. Although Black's data appeared to support Wegener's hypothesis, his samples had not been thermally treated, whereas thermal demagnetization of our samples led to isolation at 665° C of a very stable component misaligned with the NRM direction. The 665° C component probably was acquired through a C(hemical)RM and/or D(epositional)RM process by the time the rocks became consolidated. After structural correction, this component corresponds to a reverse paleomagnetic pole at 29° N,167° E ($dp = 3°$, $dm = 5°$, 9 sites). Bradore sandstone from two sites located 50 km west of Canada Bay gave comparable (virtual) poles. These results are useful for a potential test of a post-Lower Cambrian rotation of western Newfoundland.

INTRODUCTION

Newfoundland lies at the northern extremity of the Appalachian structural province and exhibits a complete cross-section of the Paleozoic—Precambrian orogen. This orogen is composed of rocks which contrast sharply with the cover rocks of the adjoining Canadian Shield in thickness, facies and structural style. Williams (1964, 1969) classified the Newfoundland orogen into three main tectonic provinces, each characterized by a distinct geologic history: the Western Platform, the Central Mobile Belt and the Avalon Platform (Fig. 1a). Based largely on the occurrence of contrasting Pacific and Atlantic faunal realms, Wilson (1966) proposed that the present Western and

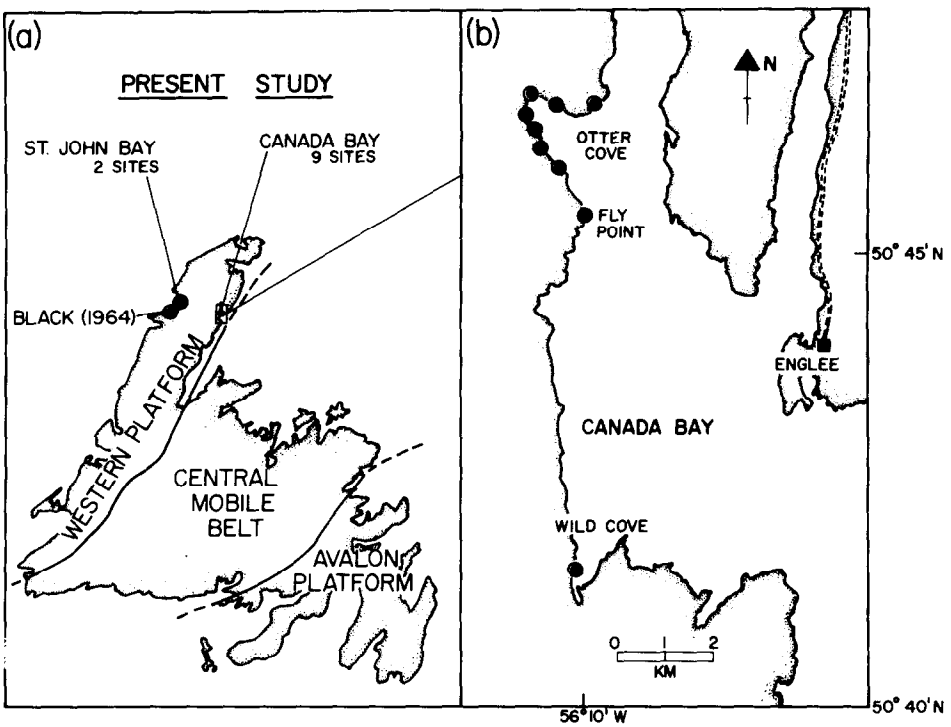


Fig. 1. (a) Paleomagnetic sampling localities in the Bradore Formation. (b) Sampling sites in Canada Bay area.

Avalon Platforms of Newfoundland had been separated by a proto-Atlantic Ocean which closed in the Lower Paleozoic and brought eastern Newfoundland into its present position relative to North America. Williams et al. (1972, 1974) further subdivided the three tectonic provinces into eight zones, each characterized by a distinct Ordovician and/or pre-Ordovician history of deposition and folding. The westernmost of these zones, called zone A (Lomond Zone) exhibits a Grenvillian continental basement unconformably overlain by a Lower Paleozoic autochthonous succession, which in turn is covered by a stack of transported slices of Lower Paleozoic sedimentary and ophiolitic sequences. Williams and Stevens (1974) have summarized the evidence confirming that zone A belongs to the ancient continental margin of eastern North America. In this paper we present the results of a paleomagnetic investigation of the Lower Cambrian Bradore sandstones in the Canada Bay area (Fig. 1), which comprise one of the autochthonous units of zone A.

Paleomagnetic results from the Bradore Formation were previously reported by Black (1964), who collected ten samples from the Highlands of St. John and used the results in comparisons with other Paleozoic paleomagnetic data from Newfoundland and mainland eastern Canada as evidence on the

hypothesis of rotation of Newfoundland; this is discussed at greater length later. Apart from Black's study, no Cambrian magnetic results on Newfoundland rocks have been previously published, and indeed the record of reliable Cambrian paleomagnetic data for all of North America is still poorly established (e.g. Vincenz et al., 1975). The present study was initiated for the foregoing reasons and forms part of a broad, paleomagnetic investigation aimed at obtaining evidence on the hypothesis of a proto-Atlantic Ocean (Wilson, 1966). Four specific objectives are: (1) to extend and refine Black's Bradore study through more detailed sampling and with the use of thermal demagnetization in assessing magnetic stability; (2) to obtain a new Cambrian pole position for Newfoundland; (3) to apply the new result in constructing a path of apparent polar wandering for western Newfoundland; and (4) in re-examining the possible rotation of that block.

GEOLOGY OF THE SAMPLING AREA

Canada Bay is located on the eastern coast of the Great Northern Peninsula, where the main exposures of Bradore sediments occur at Otter Cove and Wild Cove (Fig. 1b). These sediments comprise sandstones and conglomerate dipping gently (15—25° ESE) away from the Precambrian crystalline rocks of the interior. Betz (1939) named this sedimentary unit the Cloud Mountain Formation and correlated it lithologically with the Bradore Formation of the Lower Cambrian Labrador Group (Schuchert and Dunbar, 1934), exposed separately (1) in southeastern Labrador; (2) at St. John Bay (Fig. 1a); and (3) further south on Newfoundland's west coast. The name Bradore Formation to include the Canada Bay exposures is at present widely used and is favoured here. The total Lower Cambrian sequence increases in thickness from its eroded edge on the Labrador coast to about 90 m at the Strait of Belle Isle and about 600 m on the small islands east of Canada Bay (Lilly, 1967; Tuke, 1968).

The Bradore Formation is unfossiliferous except for vertical wormtubes *scolithus linearis* (Schuchert and Dunbar, 1934), a shallow-marine fossil present in the type area of southeastern Labrador but not at Canada Bay. An early Cambrian age of the Bradore rocks, at Canada Bay, however, is suggested by the fact that these are conformably overlain by the fossiliferous late Lower Cambrian Forteau Formation. In Canada Bay the Bradore Formation consists of a locally developed basal breccia, followed by white to grey, green and light purple arkoses and sandstones, with occasional pebble conglomerate. Towards the top of the section black hematite-stained sandstones and thin hematite pebble conglomerate predominate. The total thickness of the Bradore strata ranges from 50 to 70 m; for example, at Otter Cove the type section consists of 48 m of medium-grained, purple cross-bedded sandstones overlain by 15 m of black hematitic sandstones.

The following observations are due to Smyth (1973): the sandstones plot petrographically in the arkose, sub-arkose and orthoquartzite field, as de-

fined by Pettijohn (1957). The hematite grains are commonly well rounded to subangular and form up to 25% of the detritus, and the quartz grains commonly have a black iron oxide coating.

The observation (e.g. summary by Williams and Stevens, 1974) that vast carbonate banks lie along the entire western margin of the Appalachians, together with the fact that the Bradore sandstones in Newfoundland and Labrador are overlain by rocks conforming to Pettijohn's (1957) orthoquartzite—carbonate association, suggest that the rocks in the present study originated in a tidal environment bordering a shallow shelf sea.

The paleomagnetic interpretation will require a knowledge of the degree of severity and age of deformation of the Bradore sediments. Although much of zone A witnessed both subsidence and uplift during the deformations associated with the Late Precambrian—Cambrian orogenic cycle and the more widespread Ordovician—Devonian cycle (Williams et al., 1974), the autochthonous Bradore unit remained stable until subsidence began with the Late Cambrian. The subsequent uplift of the autochthon along with the Long Range Complex probably occurred over a long interval between Late Ordovician and Devonian time and the presence of the only major fracture in the study area, viz. the Wild Cove fault, suggesting that the final movements of this uplift took place as late as post-Acadian (Devonian) time (Smyth, 1973). However, it is uncertain from geologic considerations whether the gentle (15—20°) tilting associated with the Bradore sediments in Canada Bay should be attributed to either the Taconic (pre-Mid-Ordovician) orogeny or the Acadian orogeny or both. The problem of determining whether the stable remanence in these rocks was acquired before or after tilting is discussed later.

REMANENT MAGNETISM

Sampling and measurement

The collection was made by the authors during two field seasons devoted to sampling of Lower Paleozoic rocks in Newfoundland. Fifty-four hand samples of Bradore sandstone, oriented by Brunton compass, were collected at nine sites spanning 70 m on the shore of Canada Bay (Fig. 1b). A smaller collection, comprising thirteen reddish-brown, fine to medium-grained sandstones of the same formation, was made at two sites near Castor River, St. John Bay, about 50 km west of Canada Bay (Fig. 1a). Sampling of this locality was limited through difficulty of access, and results will be quoted only for comparison with the main Canada Bay study. For each oriented sample at least two specimens 2.5 cm in diameter and 2.4 cm in length were obtained and used in the paleomagnetic measurements.

The following equipment was used (Items 1—3 only for St. John Bay samples): (1) a PAR spinner magnetometer; (2) an alternating-field (AF) demagnetizer with 3-axis tumbler (Pearce, 1967); (3) a non-inductive furnace for

stepwise thermal demagnetization (Irving et al., 1961; Deutsch and Somayajulu, 1970); (4) an AC bridge to measure volume susceptibility (K) (Pätzold, 1972); (5) a 6-inch DC Varian electromagnet for use in isothermal remanence (IRM) measurements; and (6) a Zeiss Standard Universal microscope for petrologic work. Instruments 1—3 were operated in cancelled fields. With the spinner, single measurements of natural remanence (NRM) were usually repeatable to less than ±1% in intensity and less than ±1° in direction; however, in the case of these rocks, the estimated total uncertainty in the NRM determinations (combined orientation and measurement error) is 5% in intensity and 3° in direction. AF treatment was carried out in steps up to 800 peak Oersteds, being the maximum attainable field at that time. Thermal treatment was carried out up to 800°C; residual fields in the furnace were always less than 50 γ, and standard positioning procedures were used for reducing systematic errors due to the presence of any such fields. The applied field in measuring K was 0.31 Oe, the estimated error in K being 6%. The largest electromagnet field used was 8.0 kOe.

NRM and its stability

The NRM of two specimens from each sample was measured and a sample mean direction obtained, giving each specimen unit weight. For the total of 108 specimens from Canada Bay, values of the intensity of NRM and of susceptibility were in the range $J_n = 2 \cdot 10^{-6}$ to $8 \cdot 10^{-5}$ gauss and $K = 2 \cdot 10^{-5}$ to $5 \cdot 10^{-5}$ gauss/oersted, respectively; the corresponding range in the Koenigsberger ratio Q_n is 0.1—7, with 75% of the values lying in the range 0.8—1.5, where $Q_n = J_n/KH$ and $H = 0.5$ Oe.

Site-mean results are shown in Table I and Fig. 2. The NRM directions in the two specimens from the sample tended to agree well, though this is not generally the case for different samples at a site, as shown by the commonly low mean values of the precision parameter k. At all nine Canada Bay sites, the declinations (D) are predominantly southeastwards and the inclinations (I) are intermediate to moderately steep downwards. The grouping of site-mean directions is fair (Table II), and these directions are notably misaligned with both the axial dipole direction and the present field direction. There is no smearing towards the present field. After one year's storage at random in the laboratory in about 0.5 Oe field, all specimens were remeasured, showing maximum departures of only a few degrees from the directions obtained initially. These facts show that the rocks possess an appreciable magnetic stability and probably have been little affected by the earth's field over the past few thousand years.

Unlike Canada Bay, the two site-mean directions for St. John Bay are highly discordant (Table I).

In a pilot AF study, two specimens each from the nine Canada Bay sites were demagnetized in 12 steps up to 800 peak Oe. Fig. 3a shows for three representative examples that both the intensity (J_n) and the direction of

TABLE I

Site-mean directions of magnetization for the Bradore sandstones

Canada Bay Site	*N*	NRM				665°C			
		D	*I*	*k*	α_{95}	*D*	*I*	*k*	α_{95}
1	3	166	+37	62	16	150	+17	137	11
2	14	159	+44	9	14	155	+19	210	3
3	8	184	+61	7	22	153	+9	98	6
4	3	103	+65	11	39	131	+21	160	10
5	5	160	+38	44	12	142	+18	198	6
6	11	163	+49	18	11	145	+16	95	5
7	3	143	+38	36	21	150	+21	203	9
8	4	166	+38	5	45	145	+17	144	8
9	3	63	+67	23	26	146	+23	102	12
St. John Bay		NRM				680°C			
1	6	142	+47	15	18	142	+29	95	7
2	7	260	+50	3	42	124	+21	46	9

N = number of samples measured.
D = declination, degrees east of north.
I = inclination, degrees positive downward.
k = Fisher's precision parameter.
α_{95} = semi-angle of cone of 95% confidence for the mean direction.

NRM are virtually unaffected by this treatment, except for an initial drop in intensity in a few cases (specimen 2A); clearly, the entire remanence is highly stable against fields of the order of 800 Oe, suggesting that the carrier is hematite. Fields substantially larger than 800 Oe are required to isolate any secondary high-coercivity components, and since these were not available when this study was made, the AF demagnetizations were discontinued.

After its deposition the Bradore Formation had been subjected to at least 3 km of burial under Paleozoic and later sediments and subsequent uplift in Late Paleozoic time; this process might have produced a secondary magnetization in hematite such as a burial partial thermoremanence (PTRM) (Briden, 1965) that would be resistant to AF treatment. Following the theory of magnetic viscosity (Néel, 1949; Stacey, 1963) and experimental evidence (Chamalaun, 1964; Briden, 1965; Pullaiah et al., 1975), the probable duration (~150 m.y.) of burial of the Canada Bay Bradore rocks to 3 or 4 km and at 100—150°C would be equivalent to subjecting them to a temperature of about 500°C for the duration of laboratory experiments. Thus to remove such viscous PTRM it is necessary to carry out thermal cleaning by heating the sandstone specimens to at least this temperature.

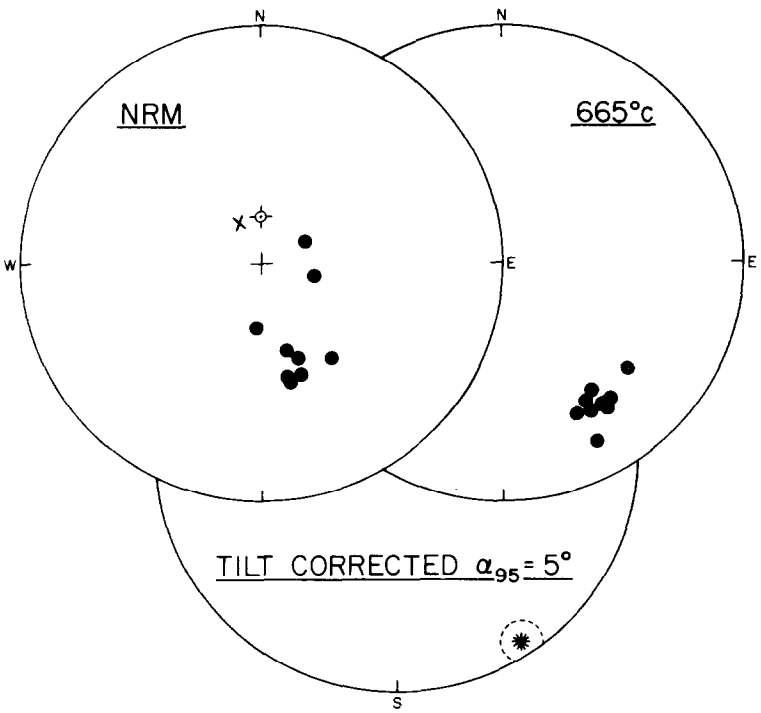


Fig. 2. Top: Site-mean directions of magnetization relative to the present horizontal, plotted on Wulff net, for the nine Bradore sites, Canada Bay; NRM and after 665°C thermal demagnetization. Dots indicate positive inclinations. -⊙- = axial dipole field, x = present field. Bottom: Mean direction (N = 9 sites) after 665°C treatment, corrected for geologic tilt, shown with circle of 95% confidence.

Two specimens from each Canada Bay site were chosen for pilot thermal demagnetizations, and three representative examples are shown in Fig. 3b. All specimens in the pilot run gave thermally discrete curves with blocking temperatures in the range 600—665°C. The magnetization decreases only slightly up to about 600°C, but is lost at temperatures close to 700°C. From the steep fall in the curves, which essentially defines the Curie temperature, it seems clear that the NRM is predominantly due to hematite. However the small decrease in intensity up to 600°C could be due to a minor amount of unaltered magnetite. Upon demagnetization the directions of all eighteen pilot samples moved progressively towards the southeast, and at 665°C became closely grouped with shallow downward inclinations. Further heating to 800°C effectively demagnetized the samples (Fig. 3b).

All remaining specimens from Canada Bay were then thermally demagnetized to 665°C, with the results summarized in Table I and Fig. 2. In the NRM the within-site directional scatter is variable and is of the same order or less than the between-site scatter (Tables I and II). At all nine sites the 665°C treatment is found to have improved the within-site grouping over

TABLE II

Area-mean directions of magnetization and corresponding pole positions

Area	Treatment	N	Mean magnetization direction				Pole position *				
			$\overline{D}$	$\overline{I}$	k	α_{95}	Lat.	Long.	dp,dm	polarity	pole no.
Canada Bay	NRM	9	153	+52	13	15	3N	146E	14,21	R	1n
	665°C	9	146	+18	106	5	23N	160E	3,5	R	1t,b
	665°C	9	144	+4	109	5	29N	167E	3,5	R	1t,a
St. John Bay	680°C	2	132	+25	—	—	13N	171E	—	R	2t,b
	680°C	2	146	+30	—	—	19N	159E	—	R	2t,a

N = number of sites used in averaging.
$\overline{D}$, $\overline{I}$ = area mean declination and inclination of remanent magnetizations.
dp,dm = semi-axes of ovals of 95% confidence.
R = reverse.
Other symbols as in Table I.

* Pole numbers are as in Fig. 5, corresponding to NRM (n) and after thermal treatment (t), and before (b) or after (a) correction for geologic tilt. Pole positions for St. John Bay are virtual geomagnetic poles.

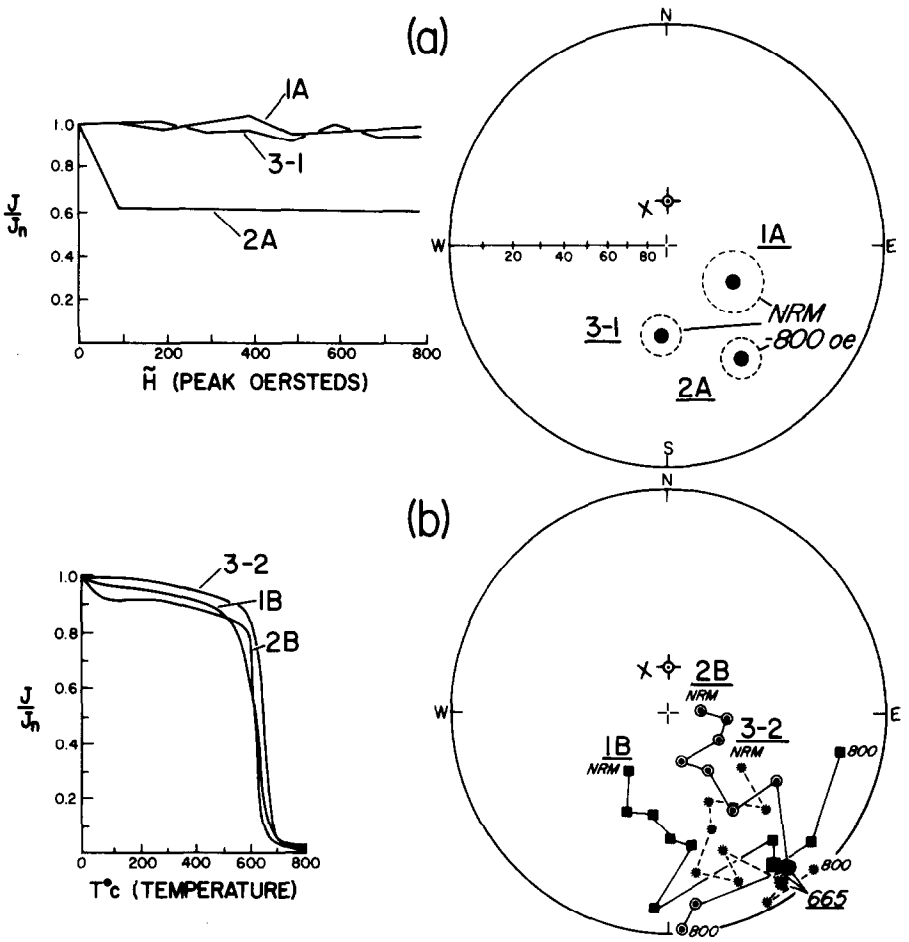


Fig. 3. Normalized intensity of magnetization J/J_n and directional changes of single Bradore specimens after stepwise (a) AF and (b) thermal demagnetization, showing three examples each from the pilot study. Numbers 1A, 1B and 2A, 2B indicate specimens from the same sample collected respectively at sites 1 and 2; 3-1, 3-2 are two different samples from the same site (3).

that of NRM. The most striking result is the production of a tight between-site grouping, $k_{665°C} = 106$, compared with $k_{NRM} = 13$, at the termination of a large directional change from the NRM.

ORIGIN OF THE MAGNETIZATION

Magnetic mineralogy

Observation by us of five polished sections under the microscope at 1000 × magnification showed hematite, typically with diameters of the order of

1 μm or less, and rarely traces of magnetite. This confirms Smyth's (1973) observations from thin sections. However, to detect possible effects due to magnetite, magnetic techniques must be used. Since the susceptibilities of the Bradore samples are generally too low to have enabled us to make Curie point determinations with a magnetic balance, we measured instead the buildup of isothermal remanence (IRM) and the remanence coercivity (H_{cr}) in fields up to 8.0 kOe, using standard techniques. In sedimentary rocks, magnetite, black specularite (>1 μm) and red pigment (<1 μm) cover roughly the ranges <1 kOe, 1—3 kOe and >3 kOe, respectively, of the coercivity spectrum (Dunlop, 1972).

Six specimens, each from a different Canada Bay site, were selected for the IRM experiments. All are relatively coarse-grained, grey sandstones with minute red staining, and their range of NRM intensities ($J_n = 10^{-6}$—10^{-4} gauss) is representative of the collection as a whole. Figure 4 shows that none of the specimens reached saturation in applied fields of 8 kOe, though the IRM acquisition curves for specimens 1, 2 and 3 have begun to level off, unlike specimens 6 (not shown), 4 and 5. The values of H_{cr} range from 3200 to 4500 Oe for all six specimens. This is characteristic for hematite in either specular or red pigment form. Another inference from this experiment is that hematite is the only component of magnetization, as no sharp inflections indicative of a magnetite/hematite association are observed in Fig. 4.

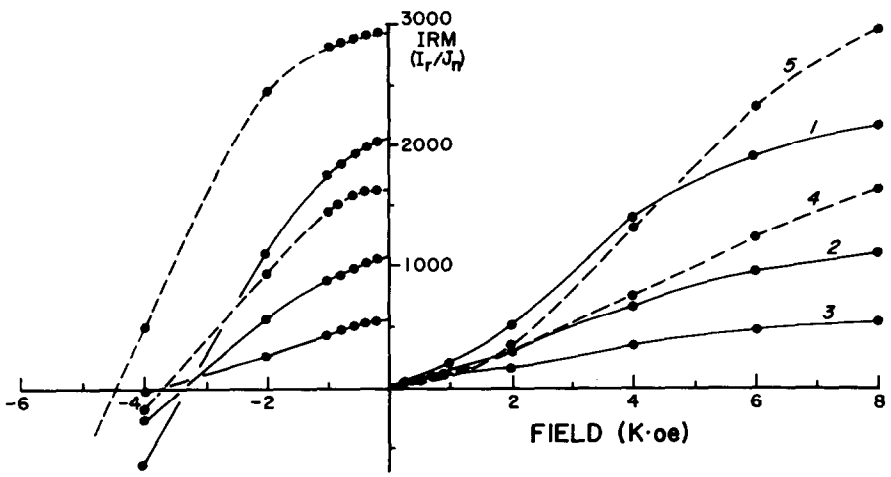


Fig. 4. Acquisition of IRM (I_r) in forward and reverse fields (right and left diagrams), normalized to the NRM intensity (J_n). Dots represent values measured in zero field after magnetization in the fields shown. The intersection of the curves with the negative field axis corresponds to the coercivity of remanence (H_{cr}). The curves are for one specimen each from Canada Bay sites 1—5; the curve from a sixth specimen (site 6) closely resembles curve 5 and is not shown. In the case of curves 4 and 5 (dashed), the actual values of I_r/J_n are twice the scale values shown.

As noted already, occasional AF curves and possibly also thermal decay curves (Fig. 3a, b; sample 2?) point to a possible small contribution from magnetite. However, this evidence is insufficient to indicate whether or not the specular hematite grains are the result of alteration of magnetite since the time of deposition.

Mode of acquisition

We concluded above that the main remanence carrier in the Canada Bay samples must be hematite. The directional changes observed upon thermal demagnetization suggest that this carrier may consist of two components both with blocking temperatures above 600° C. Hematite in a rock such as sandstone may be either primary at the time of deposition, or it could have formed during or after diagenesis. Four broad types of origin seem plausible:

Process 1. An original depositional remanence (DRM) carried by grains hematized prior to deposition.

Process 2. Deposition with magnetite settling in a DRM-like process and becoming partly or fully oxidized to hematite at some stage during diagenesis or lithification (Van Houten, 1968). This process, in which the chemical remanence (CRM) thus acquired by the hematite is aligned with the original remanence of the magnetite, has been termed a pseudo-DRM (Hargraves and Banerjee, 1973).

Process 3. Production of a CRM in hematite precipitated as red pigment from iron-bearing solutions or by dehydration of ferric oxides (Van Houten, 1968).

Process 4. Production of a stable PTRM through prolonged deep burial (Chamalaun, 1964; Irving and Opdyke, 1965; Briden, 1965) or baking occurring after, and possibly long after, lithification of the rock. In the Bradore sediments, such remagnetization may have involved either primary or secondary hematite.

The 665° C component

The intensity of the stable component isolated at 665° C in the thermal treatment is typically about 10—20% of the NRM intensity (Fig. 3b). The other component in the NRM has lower blocking temperatures and greater within-site scatter than the 665° C component (Table I) and so may be considered less stable. However, the simple shape of the thermal decay curves is consistent with the previously discussed evidence showing hematite to be the controlling magnetic mineral. Hence the stability spectra of the two components could have considerable overlap. This, coupled with the large and systematic directional change between 20° C and 665° C (Fig. 3b), suggests that the distinction between these components is not mineralogical, but concerns possibly major differences in the mode and time of acquisition of the

remanence. It is very plausible that the 665°C component was acquired earlier.

If the hematite is a product of alteration from magnetite, then the 665°C component is likely to have originated in the manner of the DRM/CRM process (no. 2, above), whereas a DRM in hematite not derived from magnetite would correspond to Process 1. A CRM by Process 3 seems less likely to account for the 665°C component, because red pigment is scarce in these sandstones. Collinson (1974) showed that in red sandstones he examined the chief contribution to a hematite DRM comes from specularite rather than pigment; the latter tends to be of post-depositional CRM origin and is generally less stable magnetically than the specularite. Thus the high stability of a DRM or pseudo-DRM in hematite usually arises from specularite in single-domain (SD) or pseudo-SD form (Hargraves and Banerjee, 1973).

Hence a DRM or pseudo-DRM residing in specularite could plausibly explain the high stability of the 665°C component. However, a simple DRM process would probably have resulted in significantly larger within-site dispersions (or smaller k-values) than those actually observed (Tables I and II). One way of reducing between-site and/or within-site dispersions, for example by smoothing of the secular variation effect, is through the integration of magnetizing processes extending over a long time during diagenesis (Roy and Robertson, 1968). This applies if the earth's field retained a single polarity during diagenesis, whereas a polarity change would tend to cause increased scatter and a smeared distribution of directions. The present results point to a single polarity.

Therefore we favour the interpretation that the portion of the hematite responsible for the 665°C component is a stable SD or PSD specularite of depositional and perhaps partly chemical origin, whose remanence had become stabilized by the end of diagenesis or lithification. This is in accord with the observations in thin sections (Smyth, 1973) and polished sections, that the hematite grains in the Canada Bay sandstones tend to have high sphericity indices, indicating that their deposition was penecontemporaneous with that of the sandstone itself. Taken in this context, we consider the 665°C component to be a "primary" remanence, with its probable time of origin before the end of the Lower Cambrian.

The "secondary" component

Since, from foregoing arguments, the "less stable" component in the NRM is almost certainly of later origin than the "primary" (665°C) component, we shall refer to it as "secondary". However, definite conclusions as to the mode and time of origin of this remanence appear to be more difficult to reach than in the case of the primary remanence. Some relevant criteria are: (1) the moderate to large increase in the within-site dispersions, and the order-of-magnitude increase in the between-site dispersion, of the NRM compared with the 665°C values (Tables I and II); (2) the high ($\geq$600°C) block-

ing temperatures of the total NRM; and (3) the marked difference between steep secondary inclinations (inferred from the NRM vectors) and the primary remanence. Structural deformation, for example resulting from subsidence and/or uplift, can be ruled out as an explanation for (1), because of the uniform tilts observed at the Canada Bay sites, and also because such deformation would have produced large between-site dispersions in the primary component. The high blocking temperatures (2) are compatible with a CRM and/or PTRM origin, such as Processes 3 and 4 above, whereas production of a secondary DRM is difficult to envisage. The inclination difference (3) suggests that a geologically significant time interval may separate the primary and secondary magnetizations. A post-diagenetic CRM could have been produced by oxidation of magnetite remaining from an original fraction that was either used up in part to form all the hematite carrying the 665°C remanence (Process 2), or else was associated with primary hematite at the time of deposition (Process 1). Once again, a CRM associated with pigment due to dehydration of iron oxide or precipitation appears to be less probable. A CRM acquired by any process is likely to have involved heating through possibly prolonged deep burial, thus contributing a viscous PTRM (Process 4).

Whatever the cause of the secondary remanence, the ratio of its magnitude to that of the 665°C remanence varies from sample to sample (e.g. in the three specimens of Fig. 3b, the $J_{665°C}/J_n$ ratio varies by a factor of 3) thus explaining part of the within-site dispersion in the NRM. A large systematic variation in the $J_{665°C}/J_n$ ratio between different sites would be needed to account fully for the large between-site dispersion of the NRM; alternatively or in addition, this dispersion may represent a well-preserved record of secular variation. However, Process 4 might not be compatible with the recording of secular variation components, as these would tend to be smoothed out by the production of a viscous PTRM.

In summary, the secondary remanence appears to reside in essentially the same, high-blocking temperature fraction of presumably SD or pseudo-SD specular hematite that carries the 665°C component. Its origin is probably mainly chemical, though a genesis through viscous PTRM is also possible. Conditions favouring such chemical and/or thermal processes must have existed at various times, especially during the Taconic and Acadian orogenies and in the Late Paleozoic. Thus the secondary component might have originated, perhaps over a long time interval, anywhere between the Late Cambrian and Mid-Devonian, or possibly later.

PALEOMAGNETISM

Pole positions

The site-mean directions for the 665°C component are shown in Fig. 2. Giving equal weight to these nine site-mean vectors, the overall mean direc-

tion is $D = 146°$, $I = +18°$, $\alpha_{95} = 5°$ (Table II). Inclination errors associated with deposition of sedimentary rocks, though easily produced in laboratory tests, appear to occur rarely in nature (Irving and Major, 1964; Opdyke and Henry, 1969; Kent, 1973), nor would one expect such errors to be present in a CRM in sediments (Collinson, 1965). Hence it seems justified to assume that an inclination error is absent in the direction quoted above. Furthermore, this mean vector most probably does not contain a significant component due to secular variation, partly because the 50—70 m vertical span of the samples should be sufficient to average out its effects, and perhaps also due to smoothing by diagenetic processes, as discussed earlier.

Hence the assumption of a (geocentric) axial dipole appears to be justified, and the north pole calculated on that assumption from the overall mean direction is 23°S,20°W, $dp = 3°$, $dm = 5°$. This corresponds to a geographic pole to which we assign an Early Cambrian age, based on the foregoing argument that the 665°C component is primary. From comparison with published Early Paleozoic poles for North America (Table II, Fig. 5), it seems probable that the antipole (23°N,160°E; 1t,b in Table II) rather than the pole represents the ancient geographic north pole relative to the sampling sites. The similarity of the geologic tilts at all Canada Bay sites rules out a fold test for determining whether or not the primary component was acquired before tilting. Use of pole position 1t,b (before tilt correction) then implies that the component from which it was calculated had its origin after tilting. Due to uncertainty in our knowledge, both of the length of the sandstone consolidation process and of the time of deformation, such a post-

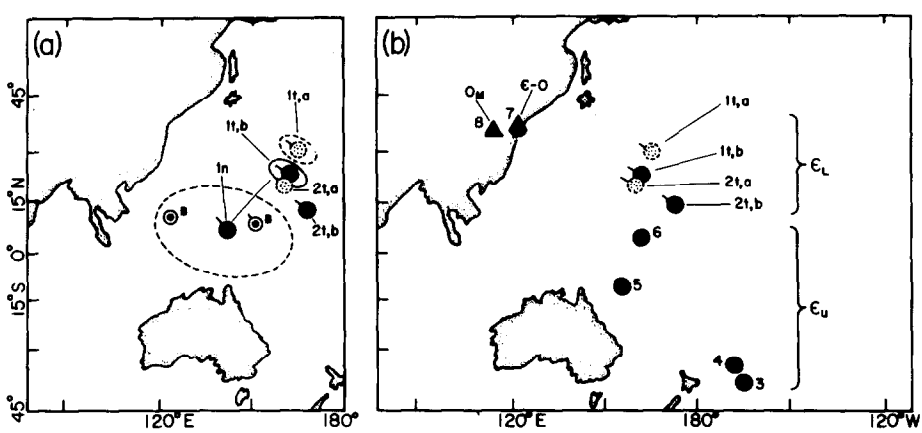


Fig. 5. (a) Pole positions relative to Bradore sandstone localities at Canada Bay (1), shown with 95% error ovals, and St. John Bay (2); n = NRM, t = after thermal treatment; b,a = before and after correction for geologic tilt. (See Table II). $\ominus^B$ and $\odot^B$ denote poles reported by Black (1964) for Bradore Formation (Fig. 1a) and Ratcliffe Brook Formation, New Brunswick.
(b) Lower Cambrian poles for western Newfoundland (1t,2t), compared with Cambrian and Ordovician poles for mainland north America (3—8).

tilting origin cannot be ruled out. However, if the 665°C component was truly "primary", then it should have preceded the deformation; in that case preference should be given to the mean direction relative to bedding (144°, +4°) and a corresponding new antipole position, which is:

29°N,167°E, $dp = 3°$, $dm = 5°$

(Table II, pole 1t,a: after tilt correction). Both poles 1t,b and 1t,a and their error ovals are shown in Fig. 5, where the use of dashed and solid outlines, respectively, reflects our preference at the time the figure was drawn, namely that the 665°C remanence originated after tilting. From the above arguments, our different view now is that the tilt-corrected pole (1t,a) probably represents the best estimate of the Lower Cambrian field direction at Canada Bay. We must stress, however, that in the case of these rocks a conclusive choice between paleomagnetic data corrected, and not corrected, for tectonism is not possible without further sampling.

Corresponding results for the St. John Bay sandstones are shown in Tables I and II. The strongly discordant NRM site-mean directions have become fairly well aligned after treatment to 680°C; this yields a virtual pole, again of reverse polarity (2t,b) shown in Fig. 5 without error oval as it is based on two sites only. The corresponding pole after tilt correction (19°N,159°E pole 2t,a), based on different dips and strikes from those at Canada Bay, differs by 12 degrees of arc from the tilt-corrected Canada Bay pole.

Rotation of western Newfoundland

A 30° anticlockwise rotation of Newfoundland implying a nearby pivot was first proposed by Wegener (1929). A positive paleomagnetic test would yield a significant difference in magnetic declination between contemporaneous rock formations in Newfoundland and in some part of mainland North America that has remained tectonically stable since the time of the proposed rotation. Nairn et al. (1959) inferred a 20° anticlockwise rotation of Newfoundland since the Carboniferous from paleomagnetic comparisons between Newfoundland and the United States. However, the significance of this result was disputed by Du Bois (1959). Black (1964) deduced a 30° anticlockwise rotation of western Newfoundland from a paleomagnetic comparison between the Bradore Formation at Highlands of St. John (Fig. 1) and the Lower Cambrian Ratcliffe Brook sandstones in New Brunswick, and from similar comparisons between Devonian and Permo-Carboniferous rocks in western Newfoundland and the mainland maritimes. He placed this rotation in mid- to Late Devonian times and argued that it would explain observed regional structures in the northeastern Appalachians.

Black's Cambrian comparison and, except for latitude, his Devonian comparison, both conform strikingly to the positive test scheme shown in Fig. 6 (a',b') and his results have been widely quoted in the literature. However, it has since become clear that Black's evidence must be regarded as inconclu-

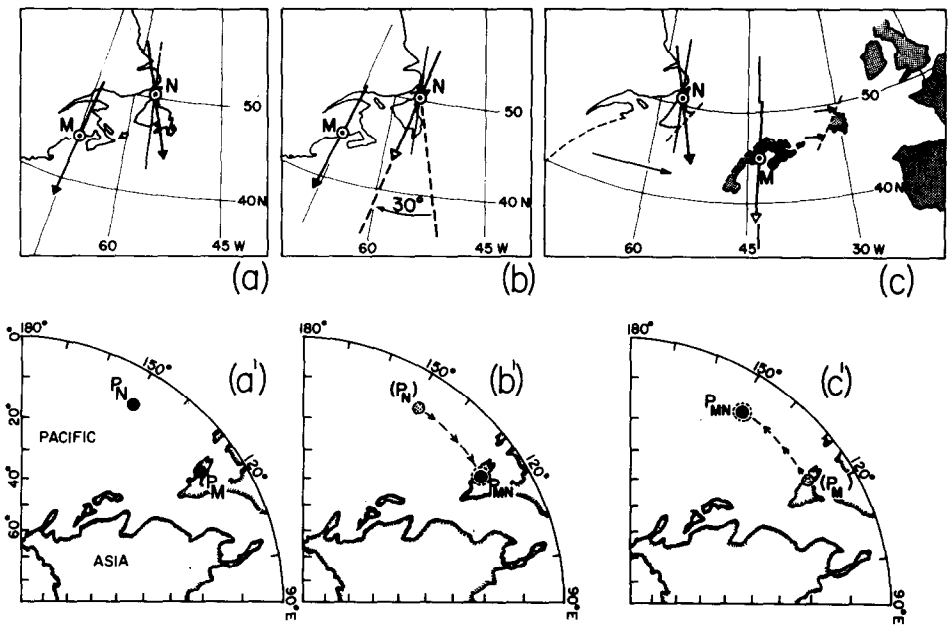


Fig. 6. Scheme illustrating the ambiguity in a paleomagnetic test of rotation. The declinations of Lower Paleozoic remanence vectors in Newfoundland (N) and the adjacent mainland (M) are depicted (a) approximately as observed (Black, 1964); (b) when Newfoundland is rotated back as shown; and (c) when instead the mainland site is rotated back to a presumed location prior to the closing of a proto-Atlantic Ocean (Wilson, 1966). Solid arrows denote the "observed" remanence vectors; hollow arrows, the "restored" vectors. Diagrams (a'), (b') and (c') show reversed pole positions corresponding to the vectors in diagrams (a), (b) and (c); P_N and P_M are "observed" poles for Newfoundland and the mainland and P_{MN} is the coincident pole obtained after rotating back Newfoundland (b') or the mainland (c'). Note that the paleomagnetic test fails to distinguish between possibilities (b) and (c) (Deutsch, 1969). Stippled portions in (c) correspond to the Atlantic faunal realm. Polar stereographic plot of the northern hemisphere.

sive. Robertson et al. (1968) cite three objections: (1) the Newfoundland and New Brunswick Cambrian remanence directions (Fig. 5a) are not different at the 95% significance level ($\alpha_{95} = 9.4°$ and 14.9°; Black, 1964); (2) the corresponding "Devonian" directions for the two areas may not be contemporaneous and they are perhaps not significantly different either; (3) AF demagnetization to 300 Oe peak fields only was carried out, and this is much too low to detect differences between high-stability components.

Figure 5a shows that the antipole position corresponding to the NRM mean direction for Canada Bay (1n, Table II) is not significantly different from either the Bradore pole or the Ratcliffe Brook pole cited by Black (1964), both being antipoles corresponding to a reverse polarity. Although the secondary component at Canada Bay may have been acquired before tilting of the strata, a post-tilting origin is more likely than in the case of the

primary component. The presence of both components in the NRM, however, makes it very difficult to relate the resultant magnetization to structure; hence we did not apply a tilt correction to the NRM direction, nor does Black report having made such a correction at his two localities. Since Black found the directions at both localities to be practically unaffected by the AF treatment, as one would expect (compare Fig. 3a), they may be considered as equivalent to NRM directions. Hence the identity of his Bradore pole with our NRM pole for Canada Bay is not surprising. Then the shift of the Canada Bay pole after thermal cleaning (both 1t,b and 1t,a) away from pole 1n and also from both of Black's poles is significant; the corresponding shift of the St. John Bay virtual poles (2t,b and 2t,a) also could be significant. This and the much improved grouping of the Canada Bay poles affords a good illustration of the value of thermal cleaning in hematite-bearing sandstones, and it confirms the validity of objection 3 by Robertson et al.

Geophysical evidence against rotation of western Newfoundland is reported by Sheridan and Drake (1968) from a seismic survey in the Gulf of St. Lawrence which failed to reveal pre-Carboniferous rifting. These authors, and also Belt (1969), suggest that Black's results might be better explained by local rotations of the New Brunswick Cambrian and Devonian rocks. An objection raised by Deutsch (1969) based on large-scale tectonics is that even if divergent Lower Paleozoic paleomagnetic results obtained from western Newfoundland and New Brunswick sites could be assumed to be perfect, they would not constitute unambiguous evidence for the rotation of Newfoundland: such results could equally well be interpreted as supporting Wilson's (1966) model for a proto-Atlantic Ocean which, in closing, transposed certain land fragments including the New Brunswick paleomagnetic sites from its original eastern margin to present localities. This is illustrated schematically in Fig. 6.

For example, whereas a new study of the Lower Cambrian Ratcliffe Brook Formation to include thermal demagnetization is clearly of interest, the results from these and any other mainland sites of presumed "European" origin are ineligible for a conclusive test of the rotation of Newfoundland. Unambiguous evidence would require data from sites in the stable interior of North America for comparison with the Newfoundland sites. An obvious first step is to look for new Lower Cambrian paleomagnetic data from the mainland for comparison with the thermally treated result for Canada Bay. However, Cambrian published results for North America include only a few rock units having a reliably Cambrian, stable remanence: an up-to-date account of this is given by Vincenz et al. (1975). The more reliable Cambrian poles are shown in Table III and Fig. 5b. For all four mainland poles (3—6), a Late Cambrian age has been cited, and poles 5 and 6 are discordant with poles 3 and 4. It is uncertain whether one would be justified in drawing a path of apparent polar wandering connecting pole pair 3,4 with the pair 5,6 and terminating at the Cambro-Ordovician and mid-Ordovician poles 7 and 8. In any case, there is at present no reliable basis for comparison between rock

TABLE III

Cambrian and Ordovician poles for Mainland N. America

Pole No. [1]	Rock unit	Age [2]	Pole position (*dp,dm*)	Polarity	Reference
3	Lamotte Sandstones, Group IIa, Missouri	Late Cambrian	38S,168W (4,8)	N	Al-khafazi and Vincenz (1971)
4	Bonneterre Dolostones, Missouri	Late Cambrian	35S,170W (—)	R	Beales et al. (1974)
5	Lamotte Sandstones, Group IIb, Missouri	Late Cambrian	12S,155E (7,11)	N	Al-khafazi and Vincenz (1971)
6	Wilberns Siltstones, Texas	Late Cambrian	6N,159E (4) [4]	R	Van der Voo and Williams (1975)
7	Colorado Intrusives [3]	Cambro-Ordovician	37N,122E (6,12)	R	Larson and Mutschler (1971)
8	Trenton Limestones, New York	Middle Ordovician	36N,114E (3,5)	R	McElhinny and Opdyke (1973)

[1] As in Fig. 5.
[2] Probable age of magnetization as cited by authors.
[3] Pole based on AF-demagnetized samples only.
[4] α_{95} value quoted by authors.
N,R = normal, reverse.
Other symbols as in Table II.

units in western Newfoundland (Lower Cambrian) and in the mainland interior (Upper Cambrian), and a synchronous Cambrian test of rotation of western Newfoundland must await new paleomagnetic data from the mainland and/or Newfoundland.

New paleomagnetic data from other than Cambrian rock formations in western Newfoundland have now become available for a conclusive test. These are: (1) the Grenville anorthosite inlier at Indian Head (Murthy and Rao, 1976); (2) the Cloud Mountain basalts (Rao, 1975), which underlie the Bradore sandstones at Canada Bay and which may be of Eo-Cambrian age; (3) Lower Ordovician limestone (Deutsch and Rao, 1975) and dolostone (Beales et al., 1974), both of the St. George Formation; and (4) Early to mid-Ordovician pillow lava and sheeted dykes at Moreton's Harbour (Deutsch et al., 1973). Although Moreton's Harbour is part of the Central Mobile Belt rather than the Western Platform of Newfoundland, it seems reasonable to group together the results under (4) and (3), as the oceanic rocks in the former case most probably formed part of the North American margin at the time of their origin (D.F. Strong, personal communication).

Deutsch and Rao (1975) recently applied these results in a re-examination of the hypothesis of rotation of Newfoundland, by comparing the pole positions with published poles from rocks of respectively Ordovician, latest Precambrian and Grenville age in the interior United States and Canada. In each case the comparison fails to support the proposed 30° rotation of western Newfoundland, though a possible rotation by 5—10 degrees cannot yet be excluded. On the other hand, the two poles of the St. George Formation (3, above) diverge widely from the pole position based on the Wabana oolitic sandstones, eastern Newfoundland (Deutsch and Rao, 1970), which are of similar age. This is compatible with the existence of an open proto-Atlantic Ocean (Fig. 5c) in Early Ordovician time.

ACKNOWLEDGMENTS

Special thanks are due to R.R. Pätzold for carrying out the IRM experiments, and to Dr. W.R. Smyth for many geologic discussions and for field assistance. We are grateful also to Dr. P.S. Rao and H.G. Dawe for help in collecting the samples; T. Vallis for rock preparation; Dr. G.S. Murthy for a petrologic examination; R.K. Stevens for valuable discussions; Dr. H.H. Bostock (Geological Survey of Canada) for detailed advice on the Bradore Formation; Memorial University of Newfoundland for a Research Assistantship to K.V. Rao; and the National Research Council of Canada who supported this research through Grant A-1946.

REFERENCES

Al-khafazi, S.A. and Vincenz, S.A., 1971. Magnetization of the Cambrian Lamotte Formation in Missouri. Geophys. J. R. Astron. Soc., 24: 175—205.

Beales, F.W., Carracedo, J.C. and Strangway, D.W., 1974. Paleomagnetism and the origin of Mississippi Valley-type ore deposits. Can. J. Earth Sci., 11: 211—223.

Belt, E.S., 1969. Newfoundland carboniferous stratigraphy and its relation to the Maritimes and Ireland. In: M. Kay (Editor), North Atlantic — Geology and Continental Drift. Am. Assoc. Pet. Geol., Mem., 12: 734—753.

Betz, F. Jr., 1939. Geology and mineral deposits of the Canada Bay area, northern Newfoundland. Newfoundland Geol. Surv. Bull., 16, 53 pp.

Black, R.F., 1964. Paleomagnetic support of the theory of rotation of the western part of the island of Newfoundland. Nature, 202: 945—948.

Briden, J.C., 1965. Ancient secondary magnetizations in rocks. J. Geophys. Res., 70: 5205—5221.

Chamalaun, F.H., 1964. Origin of the secondary magnetization of the Old Red Sandstone of the Anglo-Welsh Cuvette. J. Geophys. Res., 69: 4327—4337.

Collinson, D.W., 1965. Origin of remanent magnetization and initial susceptibility of certain red sandstones. Geophys. J. R. Astron. Soc., 9: 203—217.

Collinson, D.W., 1974. The role of pigment and specularite in the remanent magnetism of red sandstones. Geophys. J. R. Astron. Soc., 38: 253—264.

Deutsch, E.R., 1969. Paleomagnetism and North Atlantic paleogeography. In: M. Kay (Editor), North Atlantic — Geology and Continental Drift. Am. Assoc. Pet. Geol., Mem., 12: 931—954.

Deutsch, E.R. and Rao, K.V., 1970. Paleomagnetism of Ordovician sedimentary rocks from Bell Island, Newfoundland. EOS., 51: 272 (abstract).

Deutsch, E.R. and Rao, K.V., 1975. New paleomagnetic evidence fails to support rotation of Newfoundland. I.U.G.G. Symposium, Grenoble (abstract).

Deutsch, E.R. and Somayajulu, C.R., 1970. Paleomagnetism of Ordovician ignimbrites from Killary Harbour, Eire. Earth Planet. Sci. Lett., 7: 337—345.

Deutsch, E.R., Radhakrishnamurty, C., Strong, D.F., Rao, K.V., Pätzold, R.R. and Likhite, S.D., 1973. Magnetism of volcanic rocks from the Newfoundland Mobile Belt. I.A.G.A. Symposium, Kyoto (abstract).

Du Bois, P.M., 1959. Paleomagnetism and rotation of Newfoundland. Nature, 184: 63—64.

Dunlop, D.J., 1972. Magnetic mineralogy of unheated and heated red sediments by coercivity spectrum analysis. Geophys. J. R. Astron. Soc., 27: 37—55.

Hargraves, R.B. and Banerjee, S.K., 1973. Theory and nature of magnetism in rocks. In: F.A. Donath (Editor), Earth and Planetary Science. Annual Reviews Inc., Palo Alto, Calif., 1: 269—296.

Irving, E. and Major, A., 1964. Post-depositional detrital remanent magnetization in a synthetic sediment. Sedimentology, 3: 135—143.

Irving, E. and Opdyke, N.D., 1965. The paleomagnetism of the Bloomsburg red-beds and its possible application to the tectonic history of the Appalachians. Geophys. J. R. Astron. Soc., 9: 153—167.

Irving, E., Stott, P.M. and Ward, M.A., 1961. Demagnetization of igneous rocks by alternating magnetic field. Philos. Mag., 6: 225—241.

Kent, D.V., 1973. Post-depositional remanent magnetization in deep-sea sediment. Nature, 246: 32—34.

Larson, E.E. and Mutschler, F.E., 1971. Anomalous paleomagnetic pole from isotopically dated Cambro-Ordovician intrusives in Colorado. Geol. Soc. Am. Bull., 82: 1657—1666.

Lilly, H.D., 1967. Some notes on stratigraphy and structural style in central west Newfoundland. In: E.R.W. Neale and H. Williams (Editors), Geology of the Atlantic Region. Geol. Surv. Can., Spec. Pap., 4: 201—211.

McElhinny, M.W. and Opdyke, N.D., 1973. Remagnetization Hypothesis Discounted: A paleomagnetic study of the Trenton Limestone, New York State. Geol. Soc. Am. Bull., 84: 3697—3708.

Murthy, G.S. and Rao, K.V., 1976. Paleomagnetism of Steel Mountain and Indian Head anorthosites from western Newfoundland. Can. J. Earth Sci., 13: 75—83.
Nairn, A.E.M., Frost, D.V. and Light, B.G., 1959. Paleomagnetism of certain rocks from Newfoundland. Nature, 183: 596—597.
Néel, L., 1949. Théorie du traînage magnétique des ferromagnétiques aux grains fins avec applications aux terres cuites. Ann. Geophys., 5: 99—136.
Opdyke, N.D. and Henry, K.W., 1969. A test of the dipole hypothesis. Earth Planet. Sci. Lett., 6: 139—151.
Pätzold, R., 1972. High-Temperature Magnetic Susceptibility Bridge. Unpubl. thesis, Memorial University of Newfoundland.
Pearce, G.W., 1967. Design and Calibration of Apparatus for the Alternating Field Demagnetization of Rocks. Unpubl. thesis, Memorial University of Newfoundland.
Pettijohn, F.J., 1957. Sedimentary Rocks. Harper, New York, 718 pp.
Pullaiah, G., Irving, E., Buchan, K.L. and Dunlop, D.J., 1975. Magnetization changes caused by burial and uplift. Earth Planet. Sci. Lett., 28: 133—143.
Rao, K.V., 1975. Paleomagnetism of the Lower Paleozoic Bradore sandstones and the hypothesis of rotation of western Newfoundland. 9th Meet. Geol. Soc. Am., N.C. Sect., Abstr. with programs, 7: 843 (abstract).
Robertson, W.A., Roy, J.L. and Park, J.K., 1968. Magnetization of the Perry Formation of New Brunswick, and the rotation of Newfoundland. Can. J. Earth Sci., 5: 1175—1181.
Roy, J.L. and Robertson, W.A., 1968. Evidence for diagenetic remanent magnetization in the Maringouin Formation. Can. J. Earth Sci., 5: 275—285.
Schuchert, C. and Dunbar, C.O., 1934. Stratigraphy of western Newfoundland. Geol. Soc. Am., Mem., 1: 124 pp.
Sheridan, R.E. and Drake, C.L., 1968. Seaward extension of the Canadian Appalachians. Can. J. Earth Sci., 5: 337—373.
Smyth, W.R., 1973. The Stratigraphy and Structure of the Hare Bay Allochthon, Northwestern Newfoundland. Unpubl. thesis, Memorial University of Newfoundland.
Stacey, F.D., 1963. The physical theory of rock magnetism. Philos. Mag. Suppl. Adv. Phys., 12: 46—133.
Tuke, M.F., 1968. Autochthonous and allochthonous rocks in the Pistolet Bay area in northernmost Newfoundland. Can. J. Earth Sci., 5: 501—513.
Van der Voo, R. and Williams, D.W., 1975. Paleomagnetism of the Cambrian Wilberns Formation (Texas), and the Cambrian and Hadrynian poles for the North American Continent. EOS, Trans. Am. Geophys. Union., 56: 596 (abstract).
Van Houten, F.B., 1968. Iron oxides in red beds. Geol. Soc. Am. Bull., 79: 399—416.
Vincenz, S.A., Yaskawa, K. and Ade-Hall, J.M., 1975. Origin of the magnetization of the Wichita Mountains Granites, Oklahoma. Geophys. J. R. Astron. Soc., 42: 21—48.
Wegener, A., 1929. The origin of continents and oceans. (Trans). Dover Publications, New York, translation 1966.
Williams, H., 1964. The Appalachians in northeastern Newfoundland: A two-sided symmetrical system. Am. J. Sci., 262: 1137—1158.
Williams, H., 1969. Pre-Carboniferous development of Newfoundland Appalachians. In: M. Kay (Editor), North Atlantic — Geology and Continental drift. Am. Assoc. Pet. Geol., Mem., 12: 32—58.
Williams, H. and Stevens, R.K., 1974. The ancient continental margin of eastern North America. In: C.A. Burk and C.L. Drake (Editors), The Geology of Continental Margins. Springer, New York; p. 781—796.
Williams, H., Kennedy, M.J. and Neale, E.R.W., 1972. The Appalachian structural province. In: R.A. Price and R.J.W. Douglas (Editors), Variations in Tectonic Styles in Canada. Geol. Assoc. Can., Spec. Pap., 11: 181—261.
Williams, H., Kennedy, M.J. and Neale, E.R.W., 1974. The northeastward termination of the Appalachian orogen. In: A.E.M. Nairn and F.G. Stehli (Editors), The Ocean Basins and Margins, 2. Plenum, New York; p. 79—123.
Wilson, J.T., 1966. Did the Atlantic close and then reopen? Nature, 211: 676—681.